Journal of Experimental Marine Biology and Ecology 520 (2019) 151223

Contents lists available at ScienceDirect

Journal of Experimental Marine Biology and Ecology

journal homepage: www.elsevier.com/locate/jembe

Thermal heterogeneity along the migration corridors of sea turtles: Implications for climate change ecology

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ARTICLE INFO

Keywords:
Adaptive potential
Long-distance migrants
Mismatch
Mobile species
Phenology
Species response

ABSTRACT

Increased temperatures caused by anthropogenic climate change are becoming a major challenge for species. In particular, species that migrate over long distances are affected by altered climatic conditions at the various sites they frequent and en route. Here, we investigated whether climatic conditions experienced by sea turtles during their migration would change in the future. We focused on the post-nesting migrations of female loggerhead sea turtles (*Caretta caretta*) in the Mediterranean region, to explore climatic variability along four main migration corridors and their surrounding environment. We also modeled the impact of earlier onset of migration, as a potential adaptive behavioral response of Mediterranean loggerheads. We found that over the next 80 years, sea turtles will be exposed to significant increases in sea surface temperatures (SSTs) along their key migration corridors and the surrounding area (mean rate of 0.24 °C increase per decade). The projected increase in SST falls outside the range of SSTs currently reported within the examined corridors, but falls below the threshold defined as the upper thermal limit for this species. A highly variable climatic mosaic was identified within the distinct segments of the corridors. We found no evidence to support that phenological shifts might result in turtles being exposed to climatic conditions that are similar to current levels, in the future. Our outputs offer the first comprehensive spatial assessment of climatic patterns along extended areas used for the migration of sea turtles and highlight the need to quantify the relationship between altered conditions and species physiological and energetic status. This regional analysis could provide a background methodological framework for mapping and assessing spatiotemporal patterns of climatic variability along migratory areas of marine megafauna.

1. Introduction

The rapidly changing climate means that species must develop adaptive responses fast enough to ensure a window of favorable conditions for their activities, or go extinct (Visser and Both, 2005; Bellard et al., 2012). This challenge is particularly important for migratory species, while even though they are mobile and they might be expected to track changes, they are affected by altered climatic conditions at the various sites they frequent (e.g., breeding, foraging, wintering grounds) on an annual basis (Santangeli and Lehikoinen, 2017). For such species, long distance travel is an enormous physical task, with high energy demands (Hays and Scott, 2013), which could further increase the temporal and spatial window that individuals are exposed to changing climatic characteristics. Therefore, it is important to explore the dynamics of climatic conditions during migration to evaluate the full range of potential impacts of climate change on mobile species (Ahola et al., 2004; Both, 2010).

Migratory marine species are prone to altered climatic conditions in oceanic habitats, with increased temperatures also altering their distributional range (e.g., whale shark; Sequeira et al., 2014) and triggering behavioral responses such as phenological shifts (e.g., beluga whales; Hauser et al., 2017). In addition, variability in ocean temperatures causes changes in migration routes (e.g., elephant seal; Bestley et al., 2012), with pelagic marine species migrating through alternative habitats to capture favorable conditions (Beaugrand and Kirby, 2018). Ocean temperature is also tightly related to the physiological responses and energy expenditure, especially for marine ectothermic species (Williard, 2013), with low metabolic rate allowing the efficient use of their energy reserves, particularly during long periods of migration (Hays et al., 2014). Yet, most studies for migratory species continue to focus on climatic effects at departure and arrival destinations (e.g., Muhling et al., 2011; Hauser et al., 2017) or at intermediate sites (i.e., stopover sites, Dias et al., 2010; Studds et al., 2017) during migration.

The ability to quantify how environmental conditions vary during the course of migration is critical towards assessing species fitness and

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https://doi.org/10.1016/j.jembe.2019.151223
Received 21 February 2019; Received in revised form 20 August 2019; Accepted 21 August 2019
Available online 06 September 2019
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performance (Anderson et al., 2013). At the same time, such information could inform us of whether phenological shifts (i.e., changes on the timing of seasonal biological events; Walther et al., 2002) are efficient mechanisms for coping with climate change at different periods of the life cycle. For example, an onset of the seaward migration of young anadromous Atlantic salmon (*Salmo salar*), driven by environmental cues in the freshwater habitat (e.g., water temperature, flow, turbidity), leads to adverse conditions in the marine environment (predators, pathogens, feeding opportunities, ocean temperatures), posing serious limitations to individuals' performance (Otero et al., 2014). Under this context, phenological responses that are provoked by cues (e.g., energy reserves, climate) perceived at sites located far from the destination could result in maladaptation, such as asynchrony with trophic resources or anticipation of unfavorable weather conditions (Thackeray et al., 2010; Åkesson et al., 2017). Therefore, it is important to understand whether potential changes to the initiation of migration will ensure that favorable climatic conditions are maintained throughout the migration route.

Sea turtles represent an ideal model for exploring the dynamics of climate change during migration. They have a complex life cycle with distinct foraging and breeding grounds that may be distant (up to 2800 km for hard-shelled sea turtles; Hays and Scott, 2013). Research on the potential impacts of sea turtles to climate change has focused on various aspects at breeding (e.g., Poloczanska et al., 2009; Witt et al., 2010), and foraging sites (e.g., Hazen et al., 2013; Pikesley et al., 2015). There is also evidence that an earlier onset of nesting could serve as an effective mechanism in order for sea turtles to capture favorable climatic conditions during breeding period (Almpanidou et al., 2016, 2018; Rees et al., 2016). Still, while information on migration routes and main corridors is increasing (Hays et al., 2016), there is a knowledge gap on the dynamics of climatic conditions during sea turtle migration.

Here, we investigated whether sea surface temperature (SST) experienced by sea turtles during their migration would change in the future. We focused on the post-nesting migrations (i.e., from breeding to foraging grounds) of female loggerhead sea turtles (*Caretta caretta*) in the Mediterranean region. We developed a dynamic multifaceted process to assess the potential exposure of individuals to variability in SST, by successively comparing the current climatic properties of segments of corridors and the surrounding area, considered as “baseline conditions”, to projected conditions, while animals move in the seascape. Next, to examined whether variation in climatic conditions confronted during post-nesting migrations would be limited by an earlier onset of migration, we projected future SSTs considering an advance in the initiation of migration. The proposed approach could be applied for modeling climatic exposure along other critical migration travels (e.g., post hatchling, adult pre-breeding, juvenile developmental migrations), but also for other highly migratory marine species. Thus, this spatio-temporal framework could be used to advance our understanding on the impacts of climate change during migration, which is a highly demanding physical task.

2. Material and methods

2.1. Migration tracks

To determine the spatial distribution of post-nesting migration corridors of loggerhead sea turtles in the Mediterranean region, we compiled a database of all available satellite tracked data. We initially searched Google Scholar using the terms “sea turtles” (or “marine turtles”), “satellite telemetry” and “Mediterranean”. In the search, we included scientific papers and grey literature (i.e., conference proceedings, theses and reports). Since 1982 that the first reference for tracking sea turtles by satellite was made (Timko and Kolz, 1982), a total of 916 records were identified (by 20 December 2017). In addition, we validated our research against the sources provided by Luschi and Casale (2014), adding more recent publications.

Sources that were for species other than loggerheads or described part of movement other than the route from breeding to foraging grounds of adult animals, were excluded. Similarly, as female and male sea turtles demonstrate different temporal patterns of departure from breeding sites (i.e., in the Mediterranean region, males depart from nesting sites after mating in early summer, while females depart a few months later after finishing nesting; Schofield et al., 2010, 2013), we focused our analyses on post-nesting females only.

From the sources identified (see Appendix A), we extracted all the maps that illustrated migration routes. These maps were georeferenced and migration routes were digitized ($n = 89$), using ArcGIS (ESRI, 2011).

2.2. Delineating migration corridors

Four main migration corridors leading to key foraging sites for adult loggerheads in the Mediterranean were identified, reflecting > 80% of collected tracks. To define the spatial extent of each migration corridor, we applied a line kernel density estimate (KDE) approach (Steiniger and Hunter, 2013; Demšar et al., 2015) by using the tracking data as set of distinct lines. In the KDE algorithm, the selection of density parameters determines the final output (Demšar et al., 2015). Hence, to avoid any such bias in determining the width of the corridors, we used three different smoothing parameters (i.e., search radius distance known as bandwidths): a) the average of the maximum distance between different pairs of routes that had the same origin and destination, in the same year (100 km; Broderick et al., 2007; Zbinden et al., 2011), b) the average travel distance per day (Steiniger and Hunter, 2013), based on speed data reported for the studied loggerheads (i.e., mean speed of 40 km/day; e.g., Zbinden et al., 2011; Patel et al., 2015), and c) the mean of the two values generated by the previous methods (70 km).

A $0.0625^{\circ} \times 0.0625^{\circ}$ (approximately 7 km × 7 km) cell size was applied for the KDE analysis, to ensure agreement with the resolution of the climatic data used (see Section 2.3). For reproduction, sea turtles exhibit high fidelity to nesting areas (or beaches), inhabiting the marine area around these sites between each nesting event (termed internesting period; with usually 2–3 nests/female/season in the Mediterranean; Poloczanska et al., 2009). Therefore, many animals initiate their movement from similar locations, with a high density of tracking data being expected during the initial phase of migration, before animals disperse more broadly. Hence, to ensure that KDE outputs were not be driven by higher data densities in these areas, the selection of less strict percentages for the KDE distribution was applied. Therefore, the volume of the KDE distribution was determined at 75%, which is considered a more representative threshold for studying broader movement patterns (Pendoley et al., 2014). The analysis was performed using Geospatial Modeling Environment software (Beyer, 2012).

2.3. Climatic data

Sea turtles, as air breathing animals, tend to swim just below the surface during migration, to replenish oxygen stores regularly and avoid the energetic costs of vertical transport (Hays et al., 2001; Enstipp et al., 2016). Therefore, in order to examine climatic conditions along migration corridors, we used high spatial resolution SST data. SSTs for the Mediterranean region were derived from the Centro Euro-Mediterraneo sui Cambiamenti Climatici (CMCC) (Gualdi S., Centro Euro-Mediterraneo sui Cambiamenti Climatici, personal communication). The dataset contained historical and projected mean daily SST, covering the Mediterranean Sea for the period 1950–2100, at 0.0625° spatial resolution. Data were derived from the regional ocean-atmosphere coupled model COSMONEMO_MFS, generated by CMCC (Cavicchia et al., 2015, 2016). Future climatic data were based on projections of the Fifth Assessment Report (AR5) of the Intergovernmental Panel on Climate Change (IPCC) and emission scenario rcp45,

which represents an intermediate emission scenario (Moss et al., 2010). To extract SST time-series for the identified corridors, we generated mean daily SST values under current conditions, covering the period from 1987 to 2016. Mean daily SSTs were also calculated for 10-year intervals from 2017 to 2096 (i.e., 2017–2026, 2027–2036, 2037–2046, and so on until 2096).

2.4. Current and future climate along corridors

In the Mediterranean, the nesting season of sea turtles generally extends from mid-May to early August, with a peak in June and July (Schofield et al., 2013). After the completion of nesting, female sea turtles depart from breeding grounds and migrate to foraging grounds (Zbinden et al., 2008). Taking into account that July 20 was estimated as a mean departure date from breeding areas for most female sea turtles, after peak nesting (Zbinden et al., 2011; Schofield et al., 2013), we considered this date as the onset of migration.

Individuals migrate for up to four weeks, and experience SSTs that change both spatially and temporally. The Mediterranean region represents the northernmost edge of the distribution of loggerheads, and thus, individuals there encounter lower water temperature compared to other regions of their global range. To capture the exposure of loggerheads to this heterogeneous climatic environment, we developed and applied an approach that accounts for the patterns of climatic variability along migration corridors. This approach builds on three types of analyses.

Initially, we split each corridor into segments. To do so, we first drew a line that was equidistant between the edges of the corridor, as these were delineated based on the KDE approach (see Section 2.2). Next, this line was divided into parts of 120 km length to split the migration corridor into segments. These 120 km were used as an approximation of the distance covered by an animal over a 3-day period, assuming a constant travel speed of 40 km/day. The first segment represented a distance of 120 km from each nesting ground, and was characterized by the mean SST of July 20, 21, and 22. The second segment represented the distance of 120-240 km from the nesting ground, and was characterized by the mean SST of July 23, 24, and 25. We followed the same procedure to estimate the future SSTs for all the studied future 10-year periods, considering July 20 as the onset of migration. Sea turtles exhibit consistent movement strategies during post-nesting migration, adjusting travel speeds based on day/night, stage of migration and oceanic conditions (Dujon et al., 2017), with a wide range of reported speeds for loggerheads (e.g., from 33.6 to 45.6 km/day; Zbinden et al., 2008; from 36 to 52.8 km/day; Patel et al., 2015). Therefore, to account for this potential source of variability on the conditions experienced over different travelled distances and to alleviate the impact of daily SST variations that occur at spatial and temporal scales, we also split the corridors and averaged SST values using 5- and 7-day steps.

Of note, the different bandwidths (i.e., search radius distance) used to delineate the width of the corridors did not alter the patterns of SST ranges (see Appendix B, Fig. B.1-B.4). Thus, we maintained the migration corridors delineated with bandwidth of 70 km for the subsequent analyses, as the intermediate approach. In addition, the observed patterns in SST range of each corridor, under current and all the future periods, were similar when using 5- and 7-day steps for splitting corridors to different segments and calculating SST values (see Appendix B, Fig. B.5-B.8).

While empirical information on the optimal conditions for migration of sea turtles is not available, we compared future SSTs to SSTs currently experienced by turtles in the Mediterranean, which were considered “baseline conditions.” Thus, to investigate whether the SSTs experienced by loggerheads in the Mediterranean during their migration from breeding to foraging grounds significantly change in the future, we ran two analyses. First, we compared the distribution of current SSTs (1987–2016) in all cells of the four main corridors with the distribution of projected SSTs for each of the 10-year periods of future climate. This analysis allowed us to summarize the climatic properties experienced at the population level. Next, to investigate for potential differences in the SST range patterns among different corridors, we assessed changes to current and future SSTs for each corridor separately. The Kolmogorov-Smirnov test used to detect significant differences in SST distributions for both analyses.

To examine potential differences in SST changes for each segment of each corridor, we used a sign test. For this test, we applied a cell by cell comparison for each segment, comparing current conditions and all of the 10-year periods of future climate. To define the range of current conditions, and to avoid outliers and extreme values, we only considered cells with SSTs that fell within 95% of the distribution of all cells in each corridor.

2.5. Current and future climate around corridors

To examine differences in the potential changes of SSTs inside and outside the delineated corridors, we created buffer zones around the corridors to a distance of 70 km (i.e., to be consistent to the bandwidth used for the delineation of the corridors; see Section 2.2), wherever feasible (i.e., excluding parts of buffer zones located over the land). Following the same approach used in Section 2.4, we split each buffer zone into 120 km segments, as an approximation of the distance covered by an animal over a 3-day period. Each segment was characterized by the mean SST over three days, under current conditions, considering July 20 as the onset of migration. We followed the same procedure to estimate the future SST for all the studied future 10-year periods.

To examine potential differences regarding future SST changes between each segment of the delineated corridors and the surrounding area, we used a Mann-Whitney test. For this test, we applied a segment by segment comparison, by comparing each segment of the corridor with its adjacent sites for all of the 10-year periods of future climate. We further calculated the percentage of the cells within each segment of the adjacent sites, for which projected SST fell outside the range of the current conditions reported within the same corridor (i.e., SSTs within 95% of the distribution of all cells within each corridor).

To investigate whether a potential spatial shift of the existing corridors could ensure that future climate conditions would be similar to the current level, we assessed changes to current SSTs within each corridor and future conditions surrounding each corridor. For this analysis, we compared the distribution of current SSTs (1987–2016) by using all cells of each corridor with the distribution of projected SSTs for each 10-year period of future climate in the surrounding sites using Kolmogorov-Smirnov test.

2.6. Efficiency of phenological shifts

To investigate whether variation in climatic conditions experienced during post-nesting migrations would be limited by an earlier onset of migration, we projected future SSTs considering an advance in the onset of migration by one day per year. The rate of one day per year was recorded for the earlier onset of nesting at a major nesting site for loggerheads in the region (i.e., Zakynthos Island, Greece; Mazaris et al., 2008). Mazaris et al. (2008) also found that the duration of the nesting season did not significantly change over time. Thus, for the purposes of this study, we assumed that departure from the breeding site would shift to one day earlier too, although a number of individual life-history and environmental components might impact this behavior. Therefore, by advancing migration by one day per year, July 10 was set as the departure date for the period 2017–2026, June 30 was set as the departure date for 2027–2036 and so on.

At the population level, we examined whether phenological shifts in departure after nesting could contribute in maintaining thermal conditions by comparing current to projected distributions of SSTs along the four migration corridors. The comparisons were performed for all

10-year future periods, by applying a Kolmogorov–Smirnov test.

To assess potential changes at each corridor separately, we used a sign test to run cell by cell comparisons for each segment of the delineated corridors between current SSTs and future SSTs projected under the earlier onset of migration. We further calculated the percentage of cells within each segment, for which projected SST with phenological shifts fell outside the range of current conditions (i.e., SSTs within 95% of the distribution of all cells in each corridor).

A more strict view on the effectiveness of earlier onset of migration could support the hypothesis that, even if future SSTs fall within the distribution of the current climatic conditions, future SSTs will be at lower levels than the SSTs projected under no advancement of migration. To test this hypothesis, we applied a cell by cell comparison for projected SSTs, with and without phenological shifts, using the sign test.

3. Results

3.1. Current and future SST conditions

Comparisons of the distribution of SSTs that are currently experienced by sea turtles along the main migration corridors with projected SSTs demonstrated significant increases ($p < .05$). Over the next 80 years, future SSTs will rise by a mean rate of 0.24 °C per decade (Figs. 1 and 2). This was also the case when repeating the analysis separately for each migration corridor, with the distribution of current SSTs shifting towards higher temperatures in the near future (for all corridors and for all future decadal periods, $p < .05$).

Significant increases were obtained when performing a cell by cell comparison of current and projected SSTs within each segment of a given corridor ($p < .05$; Fig. B.9A). Still, at this finer scale, the results did not support a clear spatiotemporal pattern. For example, during the first half of the century (i.e., 2017–2056), most cells in the segments had SSTs that fell within the range of currently observed SSTs (ranging from 48.07% to 87.39% for the four corridors). However, during the second half of the century, a smaller percentage of projected SSTs fell within the current range (ranging from 4.87% to 63.11% for the four corridors), with most SSTs exceeding the values currently obtained.

The distribution of the projected SSTs in the surrounding areas of the corridors was also higher compared to current SSTs along the corridors (in all cases, for all future decadal periods, $p < .05$; Fig. B.10). For the first half of the century, these increased temperatures in many cells from the surrounding segments fell within the SSTs detected under current conditions along each entire corridor (mean percentage of cells equal to 64.53%). However, during the second half of the century, most cells in the surrounding segments had projected SSTs beyond current levels (mean percentage of cells equal to 70.72%).

Significant differences were obtained when comparing projected SSTs between segments inside and outside the same corridor ($p < .05$) for most pair-wise comparisons ($n = 532$ out of the 568 segments analyzed). These changes were heterogeneous, with 44.54% and 49.12% of the segments in the surrounding area exhibiting higher and lower temperatures, respectively, compared to the segments inside the corridors.

3.2. Efficiency of phenological shifts

Comparison of current SSTs in the corridors versus the future SSTs with an earlier onset in the migration period demonstrated significant differences ($p < .05$). The pair-wise comparison of current and future SSTs obtained under an earlier onset of migration significantly differed in most (99%) segments ($p < .05$).

For the next three decades (up to 2046), many cells of segments exceeded current SSTs (mean percentage of cells equal to 51.77%) (Fig. B·5B). About 12.94% (ranging from 6.02–20.53%) of the cells in all segments had significantly higher SST values when comparing SSTs under earlier migration versus no phenological shift in the same period ($p < .05$; Fig. 3).

From 2047 onwards, the percentage of the segments that fall within the range of SST currently detected along each corridor noticeably dropped (from 27% at 2056 to 0% from 2076 onwards). For these decades, SSTs might be much lower than those found under current conditions. At the end of the century (2086–2096), a shift in the onset of migration would lead to lower SSTs in most segments of all corridors compared to those in the range of the current SSTs.

4. Discussion

The spatiotemporal framework used in the current study delineated a highly heterogeneous climatic mosaic along and around the corridors used for the long distance, post-nesting migration of the Mediterranean loggerhead sea turtles. Future SSTs tended to be higher in all of the segments. At the end of century, the temperature exceeded that of current ranges in each corridor, but remained below the upper thermal limit for this species globally. The multi-faceted approach developed here, demonstrated that phenological shifts might not result in turtles being exposed to climatic conditions that are similar to current levels in the future along the post-nesting migration corridors.

Warmer temperatures affect all the physiological rates and biological processes of ectothermic species (Williard, 2013), with even small temperature changes having a relatively large effect on the metabolic rates of sea turtles (Marn et al., 2017). For example, higher temperatures (along with energy richer environments) could positively affect the growth, maturation, and reproduction of loggerheads (Marn et al., 2017). For temperate regions, such as the Mediterranean, where sea turtles experience lower water temperature compared to the tropics, rising temperatures could affect behavioral mechanisms, life cycle processes and activities of turtles at different life cycle stages. For example, rising temperature could reduce the length of winter dormancy (Hochscheid et al., 2007), extend the foraging period (Witt et al., 2010), and improve nesting conditions, but reduce post-hatchling survival (Maffucci et al., 2016). Our analyses showed that, during migration, loggerheads in the Mediterranean might confront higher temperatures in the future (~31 °C to the maximum), but with these temperatures being lower than the upper thermal limits and thus, they would not be threatened by hyperthermia (i.e., 37.5 °C; Heath and McGinnis, 1980; ~40 °C; Spotila et al., 1997). During the post-nesting migration, body condition is not expected to be optimal, as females do not generally forage during the breeding periods (Hays et al., 2014). Thus, the faster consumption of energy reserves, due to increased sea temperatures, might impact fitness. For example, for an adult individual with a mean body mass of ~68.5 kg (Hays et al., 1991), a mean increase of ~2.3 °C between current conditions and the end of the century (i.e., from 24.8 °C to 27.1 °C, along the corridor that connects the breeding grounds on Zakynthos, Greece, with the foraging grounds in Tunisia; Fig. 1C), could cause oxygen consumption to increase by 47% VO_2 in mL O_2/min (based on the equation provided by Hochscheid et al., 2004). Thus, this increase in temperature along the entire route could increase daily energy expenditure from 470 kJ/day to 690 kJ/day (based on equations used for loggerheads by Hatase and Tsukamoto, 2008). This demonstrates the importance of understanding how higher SSTs could impact sea turtles during migration from a physiological perspective.

Migration behavior is innate for adult sea turtles (Scott et al., 2014). Still, individuals exhibit different levels of fidelity to the same corridors during successive post-nesting travels, varying from low to high depending on oceanic or neritic migration, respectively (Broderick et al., 2007; Schofield et al., 2010). Global climate change has already led to poleward range expansion of foraging habitats of sea turtles (e.g., leatherbacks at North East Atlantic; McMahon and Hays, 2006). Thus, adult sea turtles might gradually shift their migration corridors spatially. For example, loggerheads that migrate through open ocean

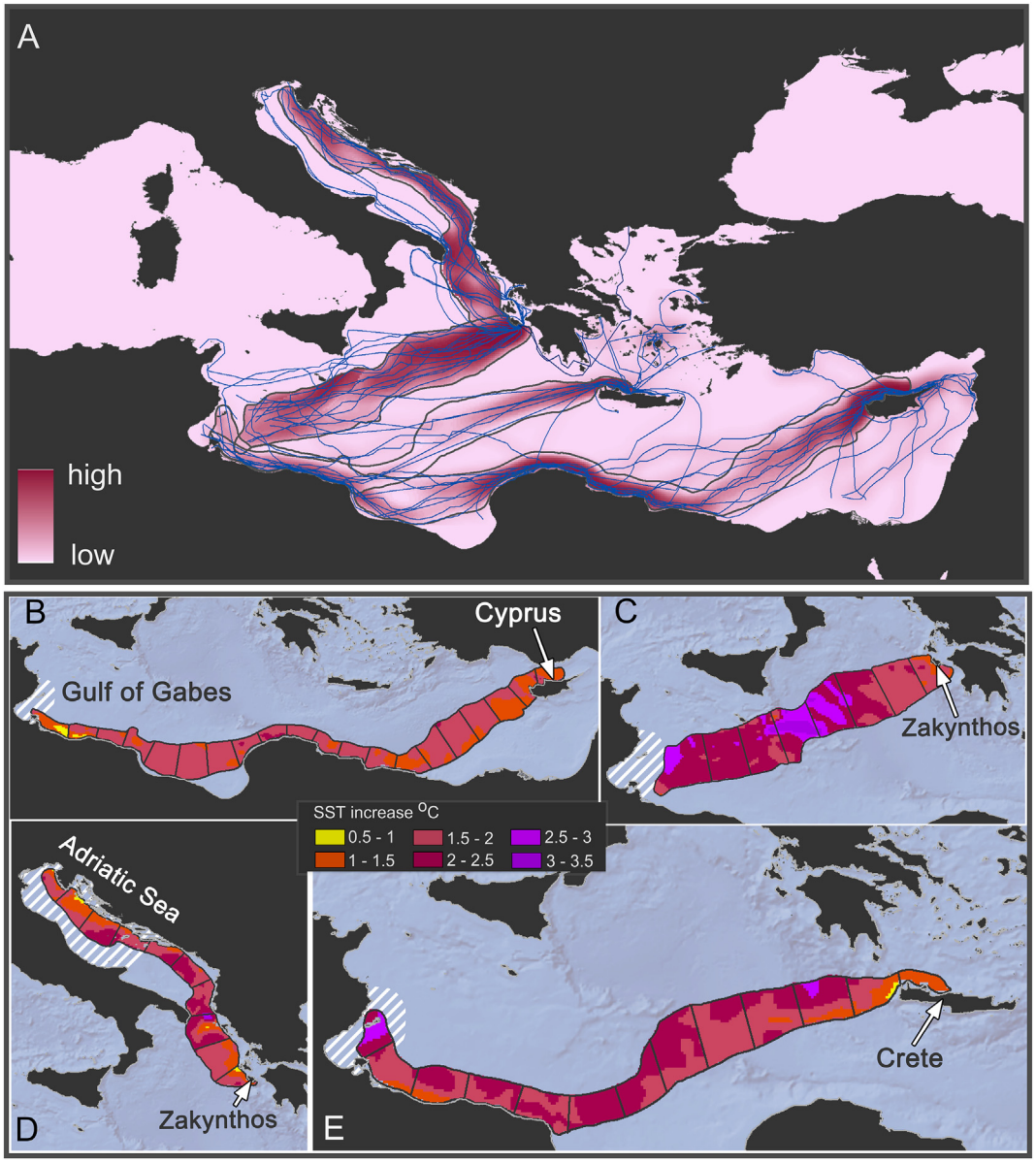


Fig. 1. The main identified post-nesting migration corridors for female loggerheads (*Caretta caretta*) in the Mediterranean region. (A) The migration routes used to determine the corridors (blue lines) and kernel density estimates of the migration tracks (red gradient) are presented. (B)–(E) Sea surface temperature increase between the current period (i.e., 1987–2016) and the end of century (i.e., 2087–2096) for the four migration corridors. Each segment of each corridor represents a 3-day step based on a mean speed of 40 km/day, departing the breeding area on 20 July. The locations of the nesting sites from which migration begins are shown. The white striped areas represent the locations of foraging areas, where migration ends up. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

habitats might move towards higher latitudes and benefit from smaller SST changes. However, probably this behavioral response could not be the case when sea turtles moving along the shore. In addition, given that our findings on the surrounding area of the main corridors revealed an heterogeneous pattern of SST change compared to the temperatures inside the corridors, it might be complicated for the animals to detect and track favorable temperatures, even if they exhibit potential spatial shifts in routes, within this highly variable climatic mosaic.

In the present study we focused on the post-nesting migration of loggerheads in the Mediterranean region, which hosts a distinct population, with specific demographic and genetic features (i.e., Regional Management Unit - RMU). The Mediterranean region represents the northernmost edge of the distribution for loggerheads, with most of breeding populations found in tropical areas where sea temperatures tend to be higher and more stable year round. Thus, this population might be subject to different evolutionary pathways to other loggerhead populations globally (Wallace et al., 2010). Still, given that climatic preferences of sea turtles are differentiated (e.g., Mazaris et al., 2015; Almpanidou et al., 2017) across RMUs, exhibiting various levels of resilience to climate change (Fuentes et al., 2013), it is important to explore potential differences in the sensitivity and the ability of distinct populations to adapt to climate change. Information on populations in warmer environments could also be used to predict behavioral future responses of populations currently inhabiting cooler habitats as these habitats warm under climate change.

To track the changing climate, sea turtles might adjust the overall travel speed based on conditions en route (e.g., SST, currents), make stopovers, or regulate the arrival to foraging areas based on climatic conditions (see Åkesson et al., 2017, for birds). For example, sea turtles have been detected to exhibit stopovers during migration at other

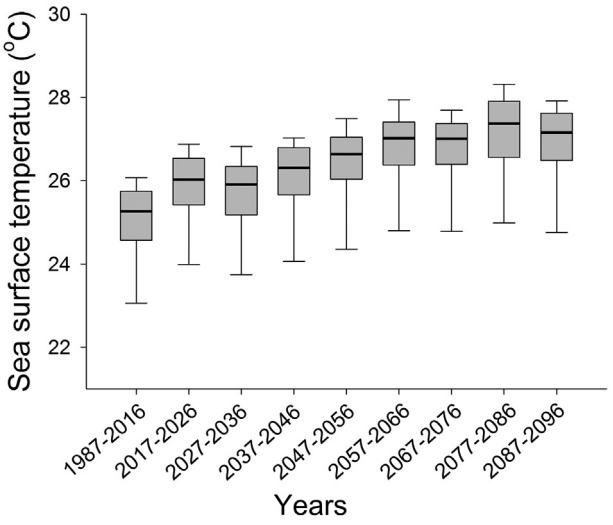


Fig. 2. Sea surface temperature along the main post-nesting migration corridors for female loggerheads (*Caretta caretta*) in the Mediterranean region over time, i.e. current 1987–2016 and future from 2017 to 2096 by 10-year periods. The thick black lines represent the median values, the upper and lower boundaries of the box represent 75th and 25th percentiles, and the vertical lines represent the 90th and 10th percentiles.

regions, primarily near the tropics (e.g., north-eastern coast of South America; Baudouin et al., 2015; Western Indian Ocean; Dujon et al., 2017) as a potential strategy for resting and feeding in order to replenish body supplies. Thus, loggerheads in temperate regions, such as the Mediterranean, might also adopt this behavior as a response to warmer temperatures in the future. For regions near the tropics, the use of cold waters at greater depth by leatherbacks (i.e., in Costa Rica) during the internesting period might be a behavioral response to avoid very warm waters and to conserve energy for reproduction (Wallace et al., 2005). In the Mediterranean region, the duration of the nesting season for loggerheads is stable through time, despite the earlier initiation (Mazaris et al., 2008); thus, the departure of turtles from the breeding grounds might initiate earlier, in parallel. However, a number of individual life history and environmental factors might influence this behavior. In other regions, shorter (Atlantic coast of central Florida, US; Pike et al., 2006; Weishampel et al., 2010) or longer (Northwest Florida, US; Lamont and Fujisaki, 2014) nesting seasons have been recorded as a response to warmer temperatures. Such an increase in temperature could result in time adjustments while animals begin their post-nesting migration. The current study showed that the climatic environment in the Mediterranean would be highly variable in the future; thus, behavioral and biological responses to warming temperatures must be monitored to determine whether they could be adopted as a population strategy in the Mediterranean.

Phenological responses do not always guarantee the synchronization of species activities with favorable conditions for growth or survival (Zipkin et al., 2012; Tomotani et al., 2018). For example, following an early departure from Africa, migratory birds might be subject to poor food supplies or adverse weather conditions at temperate breeding sites (Both, 2010; Åkesson et al., 2017). With earlier nesting, sea turtles track current climatic conditions, during nesting and incubation, which might ensure suitable habitat under climate change (Poloczanska et al., 2009). However, the current study showed that climatic conditions during the post-nesting migration might not necessarily be favorable under these phenological shifts. Thus, different life-history stages must be evaluated to elucidate the impacts of such behavioral responses.

Many other environmental factors (e.g., changes in productivity, prey distribution, ocean currents) could affect movement patterns (Putman, 2018); thus, potential future changes of these parameters could impact the observed spatiotemporal patterns. For example, there is evidence that sea turtles assess their position along their migratory routes based on geomagnetic fields (Putman and Lohmann, 2008; Brothers and Lohmann, 2015). This is also the case for other marine migrants (e.g., pink and sockeye salmon), for which geomagnetic navigation contributes to long-distance movements, along with SST and ocean currents (Putman et al., 2014). Therefore, a better understanding of the interaction between marine migratory species and their environment is needed to interpret current patterns. Such information would help improve mechanistic and hybrid modeling approaches to forecast potential species responses to climate change.

The incorporation of climate change into systematic conservation planning has largely been overlooked, despite being highlighted as a high priority (Jones et al., 2016). The proposed approach could be used as a first step towards identifying areas that would be subject to minor changes and thus, serve as climate refugia or potential favorable stepping stones through which species are likely to move in the future. However, we emphasize that further investigation is crucial to evaluate the actual impacts of these temperature changes on species performance. Under this context, we encourage future studies to apply this framework to investigate climatic variability along the migration routes used by sea turtles at the different stages of their life cycles (e.g., post hatchling, juvenile developmental, adult breeding migrations) to generate more integrated assessments.

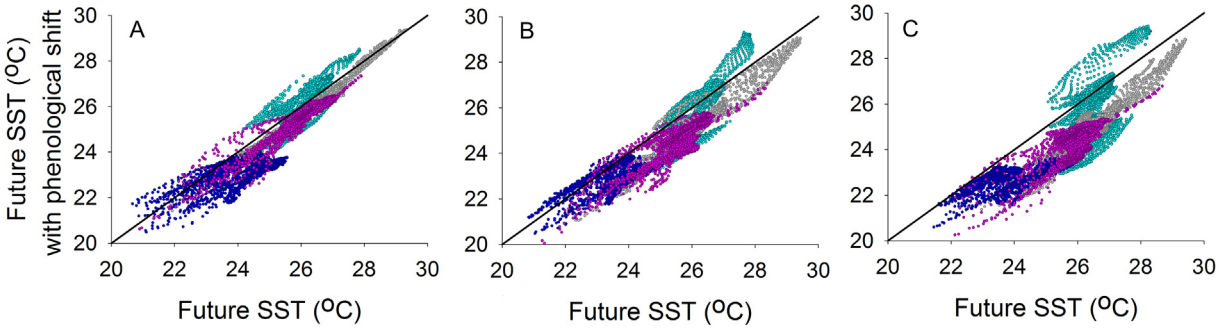


Fig. 3. Relationship between projected sea surface temperature (SST) considering an onset of migration on 20 July and projected SST accounting for phenological shifts towards an advance onset of migration by one day per year. Dots represent all the cells that comprise the main identified post-nesting migration corridors for female loggerheads (*Caretta caretta*) in the Mediterranean region for future periods of (A) 2017–2026, (B) 2027–2036 and (C) 2037–2046. Different colors represent different corridors, i.e., light blue is the corridor from the northern coast of Cyprus to Tunisia, grey is the corridor from Crete (Greece) to Tunisia, dark blue is the corridor from Zakynthos (Greece) to the northern Adriatic Sea, and pink is the corridor from Zakynthos (Greece) to Tunisia. The straight line represents an expected 1:1 relationship. The deviations from the straight line towards the upper part are indicative of cells with higher SST values, considering an earlier migration compared to SST without accounting for phenological shifts. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5. Conclusions

This study demonstrated that Mediterranean loggerheads are going to encounter increasing temperatures along and around their post-nesting migration corridors, which might impact the effectiveness of potential phenological shifts. The multifaceted approach developed and applied here could be used to model climate exposure of other highly migratory marine organisms with complex life cycles that inhabit multiple areas. We suggest that it is essential to improve our understanding of the potential impact of rising temperatures on metabolic rates, physiological responses, energetic constraints, and the overall performance of marine species, particularly ectotherms like sea turtles. Studies on climate change impacts along migration corridors should be conducted in other regions, applying the proposed methodological framework, as well as on other highly mobile marine species, to prioritize conservation actions under a changing climate.

Acknowledgements

The authors thank Dr. Silvio Gualdi from the Centro Euro-Mediterraneo sui Cambiamenti Climatici, for providing Sea Surface Temperature data for the Mediterranean Region.

Funding

This research is carried out in the context of the project “Systematic conservation planning for biodiversity: developing integrated strategies in a changing planet” (MIS 5005001) under the call for proposals “Supporting researchers with emphasis on new researchers” (EDULLL 34). The project is co-financed by Greece and the European Union (European Social Fund-ESF) by the Operational Programme Human Resources Development, Education and Lifelong Learning 2014–2020.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jembe.2019.151223.

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